弱解的L^∞估计（MOSER）*

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类似于Krylov-Safonov理论，对于散度型一致椭圆方程，建立类似的框架，即：
○ L^∞估计（对弱下解）$\Longrightarrow L^\infty$估计（对弱解）
○ Weak Harnack Inequality（L^p估计）（对弱上解）
$\Longrightarrow$ Harnack Inequality $\Longrightarrow$ C^α估计（Hölder连续）（对弱解）$\Longrightarrow$ Liouville定理.

本次讲义讨论散度型一致椭圆方程弱解的L^∞估计，方法由Moser提出，即Moser迭代法.

♣ 散度型椭圆方程.

对于区域$\Omega \subset \mathbb{R}^n$上的Laplace算子，一般有两种表示方法.

- $$\Delta u = \mathrm{tr}(D^2 u),$$

其中D^2u表示u的Hessian矩阵.

- $$\Delta u = \mathrm{div}(Du),$$

其中Du表示u的梯度.
第一种为非散度型，第二种为散度型.
对于一般的椭圆算子.

- $$Lu = \mathrm{tr}A(x)D^2u = \sum_{i,j=1}^n a_{ij}(x)D_{ij}u$$

为非散度型椭圆算子.

- $$Lu = \mathrm{div}(A(x)Du) = \sum_{i,j=1}^n D_i(a_{ij}D_ju)$$

为散度型椭圆算子.
我们讨论散度型一致椭圆方程，即.
设$\Omega \subset \mathbb{R}^n$为有界区域，$a_{ij} \in L^\infty(\Omega)$，且满足$a_{ij} = a_{ji}$，$c$在$\Omega$上可积.考虑如下算子.

$$Lu = -\sum_{i,j=1}^n D_j(a_{ij}(x)D_iu) + c(x)u \quad \text{in } \Omega.$$

*椭圆方程讨论班

且a_{ij}满足如下条件，即一致椭圆条件：

$$\lambda|\xi|^2 \le a_{ij}\xi_i\xi_j \le \Lambda|\xi|^2 \forall x \in \Omega, \xi \in \mathbb{R}^n.$$

♣ **弱解.**

定义 1 (弱解) 设$f \in L^2(\Omega)$，$u \in H^1_{\text{loc}}(\Omega)$，若$Lu \le f(\ge f)$，即对任意的$\varphi \in H^1_0$，且$\varphi \ge 0$，有

$$\int_\Omega a_{ij}D_iuD_j\varphi + cu\varphi dx \le (\ge) \int_\Omega f\varphi dx.$$

则称u为弱下解（弱上解）.特别地，若$Lu = f$，则称u为弱解.

♣ **回顾Krylov-Safonov理论的L^∞估计.**

定理 1 (L^∞估计) 设B_R为$\mathbb{R}^n$中的球，

$$Lu = a_{ij}D_{ij}u \quad in\ B_R,$$

其中a_{ij}为B_R上的连续函数，且满足

$$\lambda|\xi|^2 \le a_{ij}\xi_i\xi_j \le \Lambda|\xi|^2, \quad \forall x \in B_R, \xi \in \mathbb{R}^n.$$

设$u \in C(\bar{B}_R) \cap C^2(B_R)$满足$Lu = f$.则对任意的$p > 0$，有

$$|u|_{L^\infty(B_{R/2})} \le C\left(R^{-n/p} \parallel u \parallel_{L^p(B_R)} + R \parallel f \parallel_{L^n(B_R)}\right).$$

其中$C \sim n, p, \lambda, \Lambda$.

♣ **弱解的L^∞估计.**

定理 2 设B_R为$\mathbb{R}^n$上的球，

$$Lu = -\sum_{i,j=1}^n D_j(a_{ij}(x)D_iu) + c(x)u,$$

其中$a_{ij} \in L^\infty$，满足

$$\lambda|\xi|^2 \le a_{ij}\xi_i\xi_j \le \Lambda|\xi|^2 \quad \forall x \in B_R, \xi \in \mathbb{R}^n.$$

$c \in L^q(B_R)$，其中$q > \dfrac{n}{2}$.设$u \in H^1(B_R)$满足$Lu \le f$.若$f \in L^q(B_R)$，则$u^+ \in L^\infty_{\text{loc}}(B_R)$，且有如下估计：

$$\sup_{B_{R/2}} u^+ \le C\left(R^{-n/2} \parallel u^+ \parallel_{L^2(B_R)} + R^{2-n/q} \parallel f \parallel_{L^q(B_R)}\right).$$

$C \sim n, \lambda, \Lambda, q, R^{2-n/q} \parallel c \parallel_{L^q(B_R)}$.

证明

考虑$R = 1$的情形.

事实上，对于一般的R.

令

$$\tilde{u}(y) = u(Ry) \quad y \in B_1,$$

则$\tilde{u}$满足

$$\int_{B_1}(\tilde{a}_{ij}D_i\tilde{u}D_j\varphi+\tilde{c}\tilde{u}\varphi)dy\le\int_{B_1}\tilde{f}\varphi dy.$$

其中

$$\tilde{a}_{ij}(y)=a_{ij}(Ry),\quad \tilde{c}(y)=R^2c(Ry),\quad \tilde{f}(y)=R^2f(Ry).$$

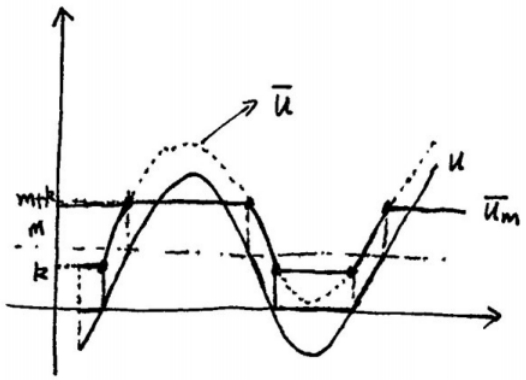
<u>$R=1.$</u>
由于u为弱解，则满足

$$\int_{B_1}a_{ij}D_iuD_j\varphi+cu\varphi dx\le\int_{B_1}f\varphi dx,$$

其中$\varphi\in H_0^1(B_1)$，且$\varphi\ge 0$.
设$k>0,m>0$，令$\bar{u}=u^++k$，

$$\bar{u}_m=\begin{cases}\bar{u} & u<m,\\ m+k & u\ge m.\end{cases}$$

则在集合$\{u<0\}\cup\{u>m\}$上$D\bar{u}_m=0$（**这里取不到端点处，是由于在$u=0,u=m$处，$\bar{u}_m$不可微.**）



当$u<m$时，显然$\bar{u}_m=\bar{u}$;当$u\ge m$时，$\bar{u}_m=m+k\le k+u=k+u^+(m>0)=\bar{u}$，从而$\bar{u}_m\le\bar{u}$.
取测试函数

$$\varphi=\eta^2(\bar{u}_m^\beta\bar{u}-k^{\beta+1})\in H_0^1(B_1),$$

其中$\beta\ge 0,\eta\in C_0^1(B_1),\eta\ge 0$.
直接计算：

$$\begin{aligned}D_j\varphi&=(\eta^2(\bar{u}_m^\beta\bar{u}-k^{\beta+1}))_j\\&=2\eta\eta_j(\bar{u}_m^\beta\bar{u}-k^{\beta+1})+\eta^2(\bar{u}_m^\beta\bar{u})_j\\&=2\eta\eta_j(\bar{u}_m^\beta\bar{u}-k^{\beta+1})+\eta^2(\beta\bar{u}_m^{\beta-1}(\bar{u}_m)_j\bar{u}+\bar{u}_m^\beta\bar{u}_j)\\&=\eta^2\bar{u}_m^\beta(\beta D_j\bar{u}_m+D_j\bar{u})+2\eta D_j\eta(\bar{u}_m^\beta\bar{u}-k^{\beta+1}).\end{aligned}$$

其中最后一个等号是由于，当$\bar{u}_m\ne\bar{u}$时，$D\bar{u}_m=0$.
由于在集合$\{u\le 0\}$中，$\varphi=0,D\varphi=0$，则下面我们在集合$\{u>0\}$中讨论.
显然当$k>0$时，$u^+\le\bar{u}$，$\bar{u}_m^\beta\bar{u}-k^{\beta+1}\le\bar{u}_m^\beta\bar{u}$.
依**Hölder不等式**，直接计算：

$$\int a_{ij}D_iuD_j\varphi=\int a_{ij}D_i\bar{u}(\eta^2\bar{u}_m^\beta(\beta D_j\bar{u}_m+D_j\bar{u})+2\eta D_j\eta(\bar{u}_m^\beta\bar{u}-k^{\beta+1}))$$

$$
\begin{aligned}
&=\int a_{ij}D_i\overline{u}(\beta D_j\overline{u}_m+D_j\overline{u})\eta^2\overline{u}_m^\beta+2\int a_{ij}D_i\overline{u}D_j\eta(\overline{u}_m^\beta\overline{u}-k^{\beta+1})\eta\\
&=\beta\int\eta^2a_{ij}D_i\bar{u}_mD_j\bar{u}_m\bar{u}_m^\beta+\int\eta^2a_{ij}D_i\bar{u}D_j\bar{u}\bar{u}_m^\beta+2\int a_{ij}D_i\bar{u}D_j\eta(\bar{u}_m^\beta\bar{u}-k^{\beta+1})\eta\\
&\geq\lambda\beta\int\eta^2\overline{u}_m^\beta|D\overline{u}_m|^2+\lambda\int\eta^2\overline{u}_m^\beta|D\overline{u}|^2-2\Lambda\int|D\overline{u}||D\eta|\overline{u}_m^\beta\overline{u}\eta\\
&\geq\lambda\beta\int\eta^2\overline{u}_m^\beta|D\overline{u}_m|^2+\lambda\int\eta^2\overline{u}_m^\beta|D\overline{u}|^2-2\Lambda\left(\int|D\overline{u}|^2\eta^2\overline{u}_m^\beta\right)^{1/2}\left(\int|D\eta|^2\overline{u}^2\overline{u}_m^\beta\right)^{1/2}\\
&=\lambda\beta\int\eta^2\overline{u}_m^\beta|D\overline{u}_m|^2+\frac{\lambda}{2}\int\eta^2\overline{u}_m^\beta|D\overline{u}|^2+\frac{\lambda}{2}\int\eta^2\overline{u}_m^\beta|D\overline{u}|^2\\
&\quad-2\Lambda\left(\int|D\overline{u}|^2\eta^2\overline{u}_m^\beta\right)^{1/2}\left(\int|D\eta|^2\overline{u}^2\overline{u}_m^\beta\right)^{1/2}+\frac{2\Lambda^2}{\lambda}\int|D\eta|^2\overline{u}_m^\beta\overline{u}^2-\frac{2\Lambda^2}{\lambda}\int|D\eta|^2\overline{u}_m^\beta\overline{u}^2\\
&=\lambda\beta\int\eta^2\overline{u}_m^\beta|D\overline{u}_m|^2+\left(\sqrt{\frac{\lambda}{2}}\left(\int\eta^2\overline{u}_m^\beta|D\overline{u}|^2\right)^{1/2}-\sqrt{\frac{2}{\lambda}}\Lambda\left(\int|D\eta|^2\overline{u}_m^\beta\overline{u}^2\right)^{1/2}\right)^2\\
&\quad+\frac{\lambda}{2}\int\eta^2\overline{u}_m^\beta|D\overline{u}|^2-\frac{2\Lambda^2}{\lambda}\int|D\eta|^2\overline{u}_m^\beta\overline{u}^2\\
&\geq\lambda\beta\int\eta^2\overline{u}_m^\beta|D\overline{u}_m|^2+\frac{\lambda}{2}\int\eta^2\overline{u}_m^\beta|D\overline{u}|^2-\frac{2\Lambda^2}{\lambda}\int|D\eta|^2\overline{u}_m^\beta\overline{u}^2.
\end{aligned}
$$

$\Longrightarrow$

$$
\begin{aligned}
\lambda\beta\int\eta^2\overline{u}_m^\beta|D\overline{u}_m|^2+\frac{\lambda}{2}\int\eta^2\overline{u}_m^\beta|D\overline{u}|^2&\leq\int a_{ij}D_iuD_j\varphi+\frac{2\Lambda^2}{\lambda}\int|D\eta|^2\overline{u}_m^\beta\overline{u}^2\\
&\leq\frac{2\Lambda^2}{\lambda}\int|D\eta|^2\overline{u}_m^\beta\overline{u}^2+\int|f|\varphi+\int|c|u\varphi\\
&\leq\frac{2\Lambda^2}{\lambda}\int|D\eta|^2\overline{u}_m^\beta\overline{u}^2+\int|f|\eta^2\overline{u}_m^\beta\overline{u}+\int|c|\overline{u}^2\eta^2\overline{u}_m^\beta
\end{aligned}
$$

$\Longrightarrow$

$$
\begin{aligned}
\beta\int\eta^2\overline{u}_m^\beta|D\overline{u}_m|^2+\int\eta^2\overline{u}_m^\beta|D\overline{u}|^2&\leq C\left(\int|D\eta|^2\overline{u}_m^\beta\overline{u}^2+\int(|c|\eta^2\overline{u}_m^\beta\overline{u}^2+|f|\eta^2\overline{u}_m^\beta\overline{u})\right)\\
&\leq C\left(\int|D\eta|^2\overline{u}_m^\beta\overline{u}^2+\int c_0\eta^2\overline{u}_m^\beta\overline{u}^2\right),
\end{aligned}
$$

其中$c_0=|c|+\dfrac{|f|}{k}$，最后一个不等式是由于$\bar{u}\geq k$.令$k=\| f \|_{L^q}$，则

$$\| c_0 \|_{L^q}\leq\| c \|_{L^q}+1.$$

设$w=\bar{u}_m^{\beta/2}\bar{u}$，则

$$Dw=\frac{\beta}{2}(\bar{u}_m)^{\beta/2}\cdot D\bar{u}_m+(\bar{u}_m)^{\beta/2}\cdot D\bar{u}$$

$\Longrightarrow$

$$|Dw|\leq\frac{\beta}{2}(\bar{u}_m)^{\beta/2}\cdot|D\bar{u}_m|+(\bar{u}_m)^{\beta/2}\cdot|D\bar{u}|$$

$\Longrightarrow$

$$
\begin{aligned}
|Dw|^2&\leq\beta^2(\overline{u}_m)^\beta\cdot|D\overline{u}_m|^2+\beta(\overline{u}_m)^\beta\cdot|D\overline{u}_m||D\overline{u}|+(\overline{u}_m)^\beta|D\overline{u}|^2\\
&\leq\beta^2(\overline{u}_m)^\beta|D\overline{u}_m|^2+\beta(\overline{u}_m)^\beta|D\overline{u}_m|^2+\beta(\overline{u}_m)^\beta|D\overline{u}|^2+(\overline{u}_m)^\beta|D\overline{u}|^2
\end{aligned}
$$

$$\leq (1+\beta)(\beta\overline{u}_m^\beta|D\overline{u}_m|^2+\overline{u}_m^\beta|D\overline{u}|^2) \quad (\beta\geq 0).$$

从而

$$|Dw|^2\eta^2 \leq (1+\beta)(\eta^2\beta\overline{u}_m^\beta|D\overline{u}_m|^2+\eta^2\overline{u}_m^\beta|D\overline{u}|^2)$$

$\Longrightarrow$

$$\begin{aligned}\int|Dw|^2\eta^2 &\leq (1+\beta)\int\eta^2\beta\overline{u}_m^\beta|D\overline{u}_m|^2+\eta^2\overline{u}_m^\beta|D\overline{u}|^2\\ &\leq (1+\beta)C\left(\int|D\eta|^2\overline{u}_m^\beta\overline{u}^2+\int c_0\eta^2\overline{u}_m^\beta\overline{u}^2\right)\\ &= C\left((1+\beta)\int w^2|D\eta|^2+(1+\beta)\int c_0w^2\eta^2\right)\end{aligned}$$

$\Longrightarrow$

$$\begin{aligned}\int|D(w\eta)|^2 &= \int|wD\eta+\eta Dw|^2\\ &\leq \int w^2|D\eta|^2+2\int w\eta|D\eta||Dw|+\int\eta^2|Dw|^2\\ &\leq \int w^2|D\eta|^2+2\left(\int w^2|D\eta|^2\right)^{1/2}\left(\int\eta^2|Dw|^2\right)^{1/2}+\int\eta^2|Dw|^2\\ &\leq \int w^2|D\eta|^2+\int w^2|D\eta|^2+\int\eta^2|Dw|^2+\int\eta^2|Dw|^2\\ &= 2\int\eta^2|Dw|^2+2\int w^2|D\eta|^2\\ &\leq C\left((1+\beta)\int w^2|D\eta|^2+(1+\beta)\int c_0w^2\eta^2\right).\end{aligned}$$

此时，上式右端第二项关于$c_0=|c|+\dfrac{|f|}{k}$，现在考虑对此项进行估计.
根据**Hölder不等式**，有

$$\int c_0w^2\eta^2 \leq \left(\int c_0^q\right)^{1/q}\left(\int(\eta w)^{\frac{2q}{q-1}}\right)^{1-1/q} \leq (\parallel c\parallel_{L^q}+1)\left(\int(\eta w)^{\frac{2q}{q-1}}\right)^{1-1/q}.$$

• $n\geq 3$.
由于

$$q>\frac{n}{2}\Longrightarrow\frac{1}{q}<\frac{2}{n}\Longrightarrow-\frac{1}{q}>-\frac{2}{n},$$

则

$$2<\frac{2q}{q-1}=\frac{2}{1-1/q}<\frac{2}{1-2/n}=\frac{2n}{n-2}.$$

选取$\theta\in(0,1)$，使得

$$\frac{2q}{q-1}=\theta\cdot\frac{2n}{n-2}+(1-\theta)\cdot 2,$$

即有

$$\theta=\frac{n-2}{2(q-1)}.$$

根据**Hölder不等式**，有

$$
\begin{aligned}
\int (\eta w)^{2q/q-1} &= \int (\eta w)^{\theta\cdot 2n/n-2+(1-\theta)\cdot 2} \\
&= \int (\eta w)^{\theta\cdot 2n/n-2}\cdot(\eta w)^{(1-\theta)\cdot 2} \\
&\le \left(\int \left((\eta w)^{\theta\cdot 2n/n-2}\right)^{1/\theta}\right)^{\theta}\left(\int \left((\eta w)^{(1-\theta)\cdot 2}\right)^{1/1-\theta}\right)^{1-\theta} \\
&= \left(\int (\eta w)^{\dfrac{2n}{n-2}}\right)^{\theta}\left(\int (\eta w)^2\right)^{1-\theta}.
\end{aligned}
$$

根据**Sobolev不等式**，有

$$
\begin{aligned}
\left(\int (\eta w)^{\dfrac{2q}{q-1}}\right)^{\dfrac{q-1}{q}} &\le \left(\int (\eta w)^{\dfrac{2n}{n-2}}\right)^{\dfrac{n-2}{2q}}\left(\int (\eta w)^2\right)^{\dfrac{2q-n}{2q}} \\
&\le C\left(\int |D(\eta w)|^2\right)^{\dfrac{n}{2q}}\left(\int (\eta w)^2\right)^{\dfrac{2q-n}{2q}}.
\end{aligned}
$$

根据**Young不等式**，有

$$
\begin{aligned}
\left(\int (\eta w)^{\dfrac{2q}{q-1}}\right)^{\dfrac{q-1}{q}} &\le C\left(\int |D(\eta w)|^2\right)^{\dfrac{n}{2q}}\left(\int (\eta w)^2\right)^{\dfrac{2q-n}{2q}} \\
&\le \varepsilon\left(\left(\int |D(\eta w)|^2\right)^{n/2q}\right)^{2q/n} + C\frac{(\varepsilon\cdot(2q/n))^{-\dfrac{2q}{2q-n}/\dfrac{2q}{n}}}{2q/(2q-n)}\left(\left(\int (\eta w)^2\right)^{(2q-n)/2q}\right)^{2q/(2q-n)} \\
&= \varepsilon\int |D(\eta w)|^2 + C\varepsilon^{-\dfrac{n}{2q-n}}\int (\eta w)^2.
\end{aligned}
$$

综合上述不等式，再根据**Young不等式**，有

$$
\begin{aligned}
\int |D(w\eta)|^2 &\le C\left((1+\beta)\int w^2|D\eta|^2 + (1+\beta)(\| c_0 \|_{L^q}+1)\left(\int (\eta w)^{2q/q-1}\right)^{1-1/q}\right) \\
&\le C(1+\beta)\int w^2|D\eta|^2 + C(1+\beta)\varepsilon\int |D(\eta w)|^2 + C(1+\beta)\varepsilon^{-\dfrac{n}{2q-n}}\int (\eta w)^2 \\
&\le C(1+\beta)\int w^2|D\eta|^2 + C(1+\beta)\varepsilon\int |D(\eta w)|^2 + C\left(\frac{2q-n}{2q}(1+\beta)^{2q/2q-n} + \frac{n}{2q}\varepsilon^{-2q/2q-n}\right)\int (\eta w)^2
\end{aligned}
$$

取ε足够小，则有

$$
\int |D(w\eta)|^2 \le C\left((1+\beta)\int w^2|D\eta|^2 + (1+\beta)^{\dfrac{2q}{2q-n}}\int w^2\eta^2\right).
$$

即有

$$
\int |D(w\eta)|^2 \le C(1+\beta)^{\alpha}\int (|D\eta|^2+\eta^2)w^2.
$$

其中$\alpha \sim n, q$.
由**Sobolev不等式**，

$$\left(\int |\eta w|^{2\chi}\right)^{1/2\chi} \leq \left(\int |D(\eta w)|^2\right)^{1/2}$$

$\Longrightarrow$

$$\left(\int |\eta w|^{2\chi}\right)^{1/\chi} \leq C(1+\beta)^{\alpha} \int (|D\eta|^2 + \eta^2) w^2,$$

其中$\chi = \dfrac{n}{n-2} > 1$.
• $n = 2$.
取$\chi > 2$，$p = \dfrac{2\chi}{1+\chi} < 2 = n$，即$\chi = \dfrac{p}{2-p}$，则

$$\begin{aligned}
\left(\int |\eta w|^{2\chi}\right)^{1/\chi} &= \left(\int (\eta w)^{2p/2-p}\right)^{\frac{2-p}{2p}\cdot 2} \\
&\leq C\left(\int |D(\eta w)|^p\right)^{2/p} \\
&\leq C\left(\left(\int |D(\eta w)^{p\cdot 2/p}\right)^{p/2} \cdot \left(\int 1^{2/2-p}\right)^{1-p/2}\right)^{2/p} \\
&\leq C\int |D(\eta w)|^2 \\
&\leq C(1+\beta)^{\alpha}\int (|D\eta|^2 + \eta^2) w^2.
\end{aligned}$$

选取截断函数如下，对任意的$0 < r < R \leq 1$，令$\eta \in C_0^1(B_R)$，且满足

$$\eta \equiv 1 \quad \text{in } B_r, |D\eta| \leq \frac{2}{R-r}.$$

从而有

$$\left(\int_{B_r} w^{2\chi}\right)^{1/\chi} \leq C\frac{(1+\beta)^{\alpha}}{(R-r)^2}\int_{B_R} w^2.$$

根据w的定义，则

$$\left(\int_{B_r} \overline{u}^{2\chi}\overline{u}_m^{\beta\chi}\right)^{1/\chi} \leq C\frac{(1+\beta)^{\alpha}}{(R-r)^2}\int_{B_R} \overline{u}^2\overline{u}_m^{\beta}.$$

令$\gamma = \beta + 2$，则

$$\left(\int_{B_r} \overline{u}_m^{\gamma\chi}\right)^{1/\chi} \leq C\frac{(\gamma-1)^{\alpha}}{(R-r)^2}\int_{B_R} \overline{u}^{\gamma}.$$

令$m \longrightarrow +\infty$，则有

$$\| \overline{u} \|_{L^{\gamma\chi}(B_r)} \leq \left(C\frac{(\gamma-1)^{\alpha}}{(R-r)^2}\right)^{1/\gamma} \| \overline{u} \|_{L^{\gamma}(B_R)},$$

现在进行迭代.
对于$i = 0, 1, 2, \cdots$，设

$$\gamma_i = 2\chi^i, \quad r_i = \frac{1}{2} + \frac{1}{2^{i+1}}.$$

由于$\gamma_i = \chi\gamma_{i-1}$，$r_{i-1} - r_i = \dfrac{1}{2^{i+1}}$，则有

$$\| \overline{u} \|_{L^{\gamma_i}(B_{r_i})} \leq C(n, q, \lambda, \Lambda)^{i/\chi^i} \| \overline{u} \|_{L^{\gamma_{i-1}}(B_{r_{i-1}})},$$

根据迭代，则有

$$\| \overline{u} \|_{L^{\gamma_i}(B_{r_i})} \leq C^{\sum \frac{i}{\chi^i}} \| \overline{u} \|_{L^2(B_1)},$$

即

$$\left(\int_{B_{1/2}} \overline{u}^{2\chi^i} \right)^{\frac{1}{2\chi^i}} \leq C \left(\int_{B_1} \overline{u}^2 \right)^{\frac{1}{2}}.$$

令$i \longrightarrow +\infty$，有

$$\sup_{B_{1/2}} \overline{u} \leq C \| \overline{u} \|_{L^2(B_1)}, \quad \text{or} \quad \sup_{B_{1/2}} u^+ \leq C \left(\| u^+ \|_{L^2(B_1)} + k \right).$$

由于$k = \| f \|_{L^q}$，则

$$\sup_{B_{1/2}} u^+ \leq C \left(\| u^+ \|_{L^2(B_1)} + \| f \|_{L^q(B_1)} \right).$$

□

引理 1 *Let $h(t) \geq 0$ be bounded in $[\tau_0, \tau_1]$ with $\tau_0 \geq 0$.Suppose,for any $\tau_0 \leq t < s \leq \tau_1$,*

$$h(t) \leq \theta h(s) + \frac{A}{(s-t)^\alpha} + B,$$

for some $\theta \in [0,1)$ and some $A, B \geq 0$.Then,for any $\tau_0 \leq t < s \leq \tau_1$,

$$h(t) \leq C \left(\frac{A}{(s-t)^\alpha} + B \right),$$

where C is a positive constant depending only on α and θ.

证明 Fix $\tau_0 \leq t < s \leq \tau_1$.
For some $0 < \tau < 1$,we consider the sequence $\{t_i\}$ defined by

$$t_0 = t \quad \text{and} \quad t_{i+1} = t_i + (1-\tau)\tau^i (s-t).$$

Note $t_i \longrightarrow s$ as $i \longrightarrow \infty$.In fact

$$\begin{aligned}
t_{i+1} &= t_i + (1-\tau)\tau^i (s-t) \\
&= t_{i-1} + (1-\tau)\tau^{i-1}(s-t) + (1-\tau)\tau^i (s-t) \\
&= \cdots \\
&= t_0 + (s-t)(1-\tau) \sum_{j=0}^{i} \tau^j \\
&= t + (s-t)(1-\tau) \frac{1-\tau^{i+1}}{1-\tau} \longrightarrow s \quad (i \longrightarrow \infty).
\end{aligned}$$

By an iteration,we get

$$\begin{aligned}
h(t) = h(t_0) &\leq \theta h(t_1) + \frac{A}{(t_1 - t_0)^\alpha} + B \\
&\leq \theta^2 h(t_2) + \theta \left(\frac{A}{(t_2 - t_1)^\alpha} + B \right) + \frac{A}{(t_1 - t_0)^\alpha} + B \\
&\leq \cdots
\end{aligned}$$

$$
\begin{aligned}
&\leq \theta^k h(t_k) + \sum_{i=0}^{k-1} \theta^i \left(\frac{A}{(t_{i+1}-t_i)^\alpha} + B \right) \\
&= \theta^k h(t_k) + \sum_{i=0}^{k-1} \theta^i \left(\frac{A}{((1-\tau)(s-t))^\alpha} \cdot \tau^{-i\alpha} + B \right) \\
&\leq \theta^k h(t_k) + \left(\frac{A}{(1-\tau)^\alpha (s-t)^\alpha} + B \right) \sum_{i=0}^{k-1} \theta^i \cdot \tau^{-i\alpha}.
\end{aligned}
$$

Choose $\tau < 1$ such that $\theta\tau^{-\alpha} < 1$,i.e.,$\theta < \tau^\alpha < 1$.As $k \longrightarrow \infty$,we have

$$
h(t) \leq C \left(\frac{A}{(1-\tau)^\alpha} (s-t)^{-\alpha} + B \right),
$$

where C is a positive constant depending only on α and θ. □

定理 3 设B_R为$\mathbb{R}^n$上的球，

$$
Lu = -\sum_{i,j=1}^{n} D_j(a_{ij}(x)D_i u) + c(x)u,
$$

其中$a_{ij} \in L^\infty$，满足

$$
\lambda|\xi|^2 \leq a_{ij}\xi_i\xi_j \leq \Lambda|\xi|^2 \quad \forall x \in B_R, \xi \in \mathbb{R}^n.
$$

$c \in L^q(B_R)$，其中$q > \frac{n}{2}$.设$u \in H^1(B_R)$满足$Lu \leq f$.若$f \in L^q(B_R)$，则$u^+ \in L^\infty_{\text{loc}}(B_R)$.进一步，对任意的$r \in (0,R)$及$p > 0$,有如下估计：

$$
\sup_{B_r} u^+ \leq C \left((R-r)^{-n/p} \parallel u^+ \parallel_{L^p(B_R)} + R^{2-n/q} \parallel f \parallel_{L^q(B_R)} \right),
$$

$C \sim n, \lambda, \Lambda, p, q$ *and* $R^{2-n/q} \parallel c \parallel_{L^q(B_R)}$.

证明

• $p \geq 2$.

By **Theorem2** and **Hölder inequality**,we have

$$
\begin{aligned}
\sup_{B_r} u^+ = u^+(y_0) &\leq \sup_{B_{(R-r)/2}(y_0)} u^+ \\
&\leq C \left((R-r)^{-n/2} \parallel u^+ \parallel_{L^2(B_{R-r}(y_0))} + (R-r)^{2-n/q} \parallel f \parallel_{L^q(B_{R-r}(y_0))} \right) \\
&\leq C \left((R-r)^{-n/2} \left(\left(\int_{B_{R-r}(y_0)} u^{+2\cdot p/2} dx \right)^{2/p} \cdot \left(\int_{B_{R-r}(y_0)} 1 dx \right)^{1-2/p} \right)^{1/2} + R^{2-n/q} \parallel f \parallel_{L^q(B_R)} \right) \\
&\leq C \left((R-r)^{-n/q} \parallel u^+ \parallel_{L^p(B_R)} + R^{2-n/q} \parallel f \parallel_{L^q(B_R)} \right).
\end{aligned}
$$

• $p \in (0,2)$.

For any $0 < \rho < r \leq R$,by **Theorem2**,we have

$$
\begin{aligned}
\sup_{B_\rho} u^+ = u^+(y_0) &\leq \sup_{B_{(r-\rho)/2}(y_0)} u^+ \\
&\leq C \left((r-\rho)^{-n/2} \parallel u^+ \parallel_{L^2(B_r)} + r^{2-n/q} \parallel f \parallel_{L^q(B_r)} \right) \\
&\leq C \left((r-\rho)^{-n/2} \parallel u^+ \parallel_{L^2(B_r)} + R^{2-n/q} \parallel f \parallel_{L^q(B_R)} \right).
\end{aligned}
$$

For $p \in (0,2)$,we have

$$\int_{B_r} (u^+)^2 dx \leq \| u^+ \|_{L^\infty(B_r)}^{2-p} \int_{B_r} (u^+)^p dx,$$

and,hence by the **Hölder inequality**,

$$\begin{aligned}\sup_{B_\rho} u^+ &\leq C\left((r-\rho)^{-n/2}\left(\frac{2-p}{2}\left(\varepsilon(\sup_{B_r} u^+)^{2-p}\right)^{2/2-p} + \frac{p}{2}\left(\frac{1}{\varepsilon}\int_{B_r}(u^+)^p dx\right)^{2/p}\right)^{1/2} + R^{2-n/q} \| f \|_{L^q(B_R)}\right) \\ &\leq \frac{1}{2}\sup_{B_r} u^+ + C\left((r-\rho)^{-n/p} \| u^+ \|_{L^p(B_r)} + R^{2-n/q} \| f \|_{L^q(B_R)}\right).\end{aligned}$$

By **Lemma1**,we have

$$\sup_{B_\rho} u^+ \leq C\left((R-\rho)^{-n/p} \| u^+ \|_{L^p(B_R)} + R^{2-n/q} \| f \|_{L^q(B_R)}\right).$$

□

定理 4 设B_R为$\mathbb{R}^n$上的球，

$$Lu = -\sum_{i,j=1}^{n} D_j(a_{ij}(x)D_i u) + c(x)u,$$

其中$a_{ij} \in L^\infty$，满足

$$\lambda|\xi|^2 \leq a_{ij}\xi_i\xi_j \leq \Lambda|\xi|^2 \quad \forall x \in B_R, \xi \in \mathbb{R}^n.$$

$c \in L^q(B_R)$，其中$q > \frac{n}{2}$.设$u \in H^1(B_R)$满足$Lu = f$.若$f \in L^q(B_R)$，则$u \in L^\infty_{\text{loc}}(B_R)$.进一步，对任意的$r \in (0,R)$及$p > 0$,有如下估计：

$$|u|_{L^\infty(B_r)} \leq C\left((R-r)^{-n/p} \| u \|_{L^p(B_R)} + R^{2-n/q} \| f \|_{L^q(B_R)}\right),$$

$C \sim n, \lambda, \Lambda, p, qR^{2-n/q} \| c \|_{L^q(B_R)}$.

证明

Apply **Theorem3** to u,we have

$$\sup_{B_r} u^+ \leq C\left((R-r)^{-n/p} \| u^+ \|_{L^p(B_R)} + R^{2-n/q} \| f \|_{L^q(B_R)}\right).$$

Apply **Theorem3** to $-u$,we have

$$\sup_{B_r} u^- = \sup_{B_r}(-u)^+ \leq C\left((R-r)^{-n/p} \| (-u)^+ \|_{L^p(B_R)} + R^{2-n/q} \| f \|_{L^q(B_R)}\right).$$

Then,we obtain

$$|u|_{L^\infty(B_r)} \leq \sup_{B_r} u^+ + \sup_{B_r} u^- \leq C\left((R-r)^{-n/p} \| u \|_{L^p(B_R)} + R^{2-n/q} \| f \|_{L^q(B_R)}\right).$$

□

参考文献

[1] Qing Han,An Introduction to Elliptic Differential Equations,Preliminary Version.